

Behnam Panahi

School of Biomedical Engineering, University of British Columbia
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Education

Ph.D. in Biomedical Engineering, University of British Columbia , Vancouver, Canada GPA: 4.0/4.0 (95.7/100)	Sep. 2022 – Present
M.Sc. in Biomedical Engineering, Sharif University of Technology GPA: 3.88/4.0 (16.75/20)	Sep. 2018 – Aug. 2021
B.Sc. in Chemical Engineering, Sharif University of Technology GPA: 3.72/4.0 (16.60/20)	Sep. 2014 – Jun. 2018

Areas of Expertise

Bioprinting and Bioengineering, Disease Modeling, Microfluidics, Stem cells biology, Biosensors.

Honors & Awards

- Selected for the Amgen Predictive Biology Cluster under the Mitacs Accelerate program to develop translational biomedical tools, University of British Columbia (2026).
- Selected for the prestigious NSERC Collaborative Research and Training Experience (CREATE) program in 3D Printing Technology and Materials (3DPTM), University of British Columbia (2022).
- Selected for the MITACS internship program through the Stem Cell Network (SCN) Training and Education Committee (2023).
- Received SCN International Travel Award to attend the TERMIS Conference, Freiburg, Germany (2025).
- Awarded SCN Organoid Symposium and Workshop Travel Award (approx. \$6000), University of Toronto, Canada (2024).
- President's Academic Excellence Initiative PhD Award, University of British Columbia (2022).
- Ranked 2nd among more than 10,000 participants in the national entrance exam for M.Sc. in Chemical Engineering (2018).

Work Experiences

Internship, Amgen <i>Burnaby, BC, Canada</i>	Jan. 2026 – Present
• Co-developing open-source, translational biomedical tools and computational pipelines to accelerate therapeutic discovery within an exclusive academic-industry partnership.	
Internship, Axolotl Biosciences <i>Victoria, BC, Canada</i>	Dec. 2023 – Jul. 2024
• Optimized bioink formulations and performed QC for 3D bioprinted cardiac models, while collaborating with business development to gain industry-facing project management experience.	

- Managed daily laboratory operations by developing SOPs, organizing resources for new research, and supervising cell culture workflows while training students on safety and experimental protocols.

Research Experiences

HeartPrint: Personalized Cardiac Tissue Models for Improved Drug Development 2022 – Present*Ph.D. research, University of British Columbia, Vancouver, Canada*

- Engineered personalized 3D bioprinted cardiac tissue models for drug screening and disease modeling by optimizing hydrogel bioinks, printing patterns, and matrix stiffness to enhance tissue maturation.

Research Assistant, NSERC CREATE Program in 3D Printing Technology and Materials (3DPTM)

2022 – Present

Supervisor: Prof. Hongbin Li, University of British Columbia, Vancouver, Canada

- Advanced interdisciplinary 3D printing and bioink research through hands-on technical training, industry seminars, and cross-functional project management.

Design and Fabrication of Cellular Patch with Microchannels

2019 – 2021

Sharif University of Technology, Tehran, Iran

- Designed and fabricated microchanneled cardiac patches that mimic vascular networks to enhance oxygen delivery, cell viability, and regenerative potential for treating heart disease.

Skills

Leadership, Teaching, and Time Management

- Experienced lab manager and teaching assistant for multiple laboratory and engineering courses.
- Skilled in mentoring and training students, delegating tasks, and leading teams.
- Developed strong time management by balancing research, teaching, and administrative responsibilities.

Prototyping & Manufacturing; 3D Bioprinting, Microfluidics and Modelling

- Skilled in 3D bioprinting with CELLINK BIO X6 and in the creation and optimization of novel bioinks.
- Proficient in microfluidic system design and simulation, including analysis of vasculature geometry on oxygen delivery and cell viability.
- Experienced in particle delivery modelling using CFD software to study particle behaviour in biological systems.

Cell and Tissue Culture

- Proficient in hiPSC culture and creation of vascularized tissue constructs.
- Experienced in differentiating hiPSCs into cardiomyocytes (hiPSC-CM) using signalling factors and quality-control assays.
- Applied cell and tissue engineering methods to various cardiac and vascular research projects.

Computer-Aided Design (CAD)

- Skilled in SolidWorks and AutoCAD for 2D and 3D design.
- Designed geometries for bioprinting from simple rectangular constructs to complex shapes such as coronary arteries.

Laboratory Techniques; SEM, Fluorescent Microscopy, UV–Infrared

- Experienced in scanning electron microscopy (SEM) for high-resolution imaging of biological samples.
- Proficient in fluorescence microscopy for visualization of labelled cells and tissues.
- Familiar with UV–visible and infrared spectroscopy for biomolecule characterization and concentration measurements.

Publications

Panahi, Behnam, Saif Dababneh, Hosna Babini, Maks Prondzynski, Mohsen Akbari, Glen Tibbits *et al.* “Versatile hiPSC Models and Bioengineering Platforms for Investigation of Atrial Fibrosis and Fibrillation” *Cells* 15, no. 2 (2026): 187.

<https://doi.org/10.3390/cells15020187>

Aghakhanolia N, Fowler J, **Panahi Behnam**, Arman Rahmim, and Pedro L. Esquinas. "A realistic lesion insertion and reconstruction framework for task-based evaluations in clinical SPECT and PET imaging." In Medical Imaging 2026: Physics of Medical Imaging, vol. 13924, pp. 767-771. *SPIE*, 2026.

<https://doi.org/10.1117/12.3087925>

Mansoori-Kermani, Amirreza, *et al.*, including **Panahi, Behnam**. “Engineered Hyaluronic Acid-Decorated Niosomal Nanoparticles for Controlled and Targeted Delivery of Epirubicin to Treat Breast Cancer.” *Materials Today Bio* 16 (2022): 100349.

<https://doi.org/10.1016/j.mtbio.2022.100349>

Conference Presentations

Oral Presentations

- **Panahi, Behnam**. “Innovative Creation of Fibrotic Atrial Tissue via 3D Bioprinting to Model Atrial Fibrillation.” *2024 Organoid Symposium*, Toronto, Canada, 2024.
- **Panahi, Behnam**. “Optimized 3D Bioprinted Atrial Tissue Models.” *TERMIS 2025*, Freiburg, Germany, 2025.

Poster Presentations

- **Panahi, Behnam**. “Building a Human Heart Model from Pluripotent Stem Cells.” *TMM 2025*, Ottawa, Canada, 2025.
- **Panahi, Behnam**. “3D Bioprinted Human Atrial Tissue: Multiparametric Modeling of Fibrosis, Arrhythmogenesis, and Drug Responsiveness.” *ISSCR 2026*, Montreal, Canada, 2026.
- **Panahi, Behnam**. “3D Bioprinted Human Tissue with Tunable Bioink Stiffness.” *MicroTAS 2026*, Granada, Spain, 2026.

TA Experiences

Teaching Assistant, University of British Columbia

Sep. 2023 – May 2026

Courses: BMEG 102, Microfluidics (BMEG 591/400L), Mechanical Design (MECH 328), CHBE 464, PHYS 131/159/100, APSC 100/101.

Head TA, Sharif University of Technology

Oct. 2018 – Sep. 2020

Courses: Fluid Mechanics Laboratory

Hobbies

Football, volleyball, swimming, fitness, hiking. In my free time, I enjoy participating in team sports and outdoor activities to stay active and healthy.

References

Dr. Nozomu Yachie

Professor, University of British Columbia

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Dr. Maksymilian Prondzynski

Principal investigator, BC Children Hospital Research Institute, Simon Fraser University

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Dr. Glen Tibbits

Professor, BC Children Hospital Research Institute, Simon Fraser University, Adjunct Prof at SBME

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